

$$\textcircled{1} \quad T(x, y, z) = T(xe_1 + ye_2 + ze_3)$$

$$= xTe_1 + yTe_2 + zTe_3$$

$$= x(2, 1) + y(0, 1) + z(1, 1)$$

$$= (2x + z, x + y + z)$$

— (2m)

To find null space:

$$N(T) = \left\{ (x, y, z) : T(x, y, z) = (0, 0) \right\}$$

$$\left. \begin{array}{l} 2x + z = 0 \\ x + y + z = 0 \end{array} \right\} \Rightarrow \begin{array}{l} \boxed{z = -2x} \\ x + y - 2x = 0 \\ \Rightarrow \boxed{y = x} \end{array}$$

$$\therefore x = 1 \Rightarrow y = 1, z = -2$$

$$N(T) = \left\{ (a, a, -2a) : a \in \mathbb{R} \right\} \quad \text{— (2 m)}$$

$$\text{nullity} = 1 \quad \text{— (0.5 m)}$$

To find range space.

$$\text{Let } A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

The basis for $R(T)$ is $\{(1, 1), (0, 1)\}$

$$\therefore R(T) = \left\{ a(1, 1) + b(0, 1) : a, b \in \mathbb{R} \right\} = \left\{ (a, a+b) : a, b \in \mathbb{R} \right\}$$

— (2 m)

$$\text{rank}(T) = 2. \quad \text{— (0.5 m)}$$

$$2) \textcircled{a} A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

Characteristic Eqⁿ of A : $\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$

Using Cayley-Hamilton thm, we have

$$A^3 - 5A^2 + 7A - 3I = 0 \quad \text{-----} \quad (1M)$$

$$\text{Now, } A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

$$= A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + A^2 + A + I$$

$$= A^2 + A + I \quad \text{-----} \quad (1M)$$

$$= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix} \quad \text{-----} \quad (1M)$$

$$\textcircled{b} B = \begin{bmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{bmatrix}$$

Eigen values of B : $1, 4, -2$. ----- (1M)

Corresponding eigenvectors : $\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ ----- (1M)

[$\frac{1}{2}$ Mark if two eigen vectors are correct]

Corresponding orthonormal eigenvectors : $\frac{1}{3}\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \frac{1}{3}\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \frac{1}{3}\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$.

$\therefore$ Orthogonal matrix P is,

$$P = \frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ 1 & -2 & 2 \\ -2 & 1 & 2 \end{bmatrix} \quad \text{-----} \quad (1M)$$

$$\text{Then, } P^{-1}BP = P^TBP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -2 \end{bmatrix}. \quad \text{-----} \quad (1M)$$

[No marks for last step, if P is incorrect]

Q.3.

We have $y_1 = 1$ and $y_2 = \frac{x^3}{3}$. Let $y_3 = y$.

$$W(y_1, y_2, y_3) = \det \begin{bmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{bmatrix}$$

$$= \det \begin{bmatrix} 1 & x^3/3 & y \\ 0 & x^2 & y' \\ 0 & 2x & y'' \end{bmatrix}$$

$$= x^2 y'' - 2xy'.$$

$$\text{We have } W = \frac{\cos(2 \ln x)}{x}.$$

$$\therefore x^2 y'' - 2xy' = \frac{\cos(2 \ln x)}{x}. \quad (1M)$$

Put $x = e^t$. Then

$$(D_t(D_t - 1) - 2D_t)y = e^{-t} \cos 2t, \quad D_t = \frac{d}{dt}.$$

$$\Rightarrow (D_t^2 - 3D_t)y = e^{-t} \cos 2t. \quad (1M)$$

$$\text{Put } y = e^{mt} \text{ in } (D_t^2 - 3D_t)y = 0. \\ \Rightarrow m^2 - 3m = 0 \Rightarrow m = 0, 3.$$

$$\therefore \text{C.F.} = c_1 e^{0t} + c_2 e^{3t} = c_1 + c_2 x^3. \quad (2M)$$

$$\begin{aligned} \text{P.I.} &= \frac{1}{D_t(D_t - 3)} e^{-t} \cos 2t \\ &= e^{-t} \frac{1}{(D_t - 1)(D_t - 4)} \cos 2t \end{aligned}$$

$$= e^{-t} \frac{1}{D_t^2 - 5D_t + 4} \cos 2t$$

$$= e^{-t} \frac{1}{-4 - 5D_t + 4} \cos 2t$$

$$= e^{-t} \frac{1}{-5D_t} \cos 2t$$

$$= -\frac{e^{-t} \sin 2t}{10} = -\frac{1}{10} \frac{\sin(2 \ln x)}{x}$$

$$\therefore y = C_1 + C_2 x^3 - \frac{1}{10} \frac{\sin(2 \ln x)}{x} \quad (2M)$$

$$y(1) = C_1 + C_2 - \frac{1}{10} \frac{\sin(2 \ln 1)}{1} = C_1 + C_2 = 0.$$

$$y' = 3C_2 x^2 + \frac{1}{10} \frac{\sin(2 \ln x)}{x^2} - \frac{1}{10} \frac{\cos(2 \ln x)}{x} \times \frac{2}{x}$$

$$= 3C_2 x^2 + \frac{1}{10} \frac{\sin(2 \ln x)}{x^2} - \frac{1}{5} \frac{\cos(2 \ln x)}{x^2}.$$

$$\therefore y'(1) = 3C_2 + \frac{1}{10} \frac{\sin(2 \ln 1)}{1} - \frac{1}{5} \frac{\cos(2 \ln 1)}{1}$$

$$= 3C_2 + 0 - \frac{1}{5} = 3C_2 - \frac{1}{5}.$$

$$\therefore y'(1) = -\frac{1}{5} \Rightarrow C_2 = 0. \text{ Thus } C_1 = 0.$$

$$y = y_3 = -\frac{1}{10} \frac{\sin(2 \ln x)}{x} \quad (1M)$$

Question: Solve the following initial value problems.

$$xy' + 2y = -x^3 y^2 \cos x, \quad y(1) = 2.$$

[7 Marks]

Solⁿ: Observe that the given differential equation is a Bernoulli equation.

In the standard form, we have

① $y' + \left(\frac{2}{x}\right)y = (-x^2 \cos x) y^2. \quad \text{--- (1)}$

Substitute

① $u = y^{1-2} = y^{-1}$

Differentiating it with respect to x

gives

$$u' = -y^{-2} y' \quad \text{--- (2)}$$

Substituting the value of y' from (1) into (2) gives

$$\begin{aligned} \textcircled{2} \quad u' &= -\bar{y}^2 \left(-x^2 \cos x \cdot y^2 - \frac{2}{x} y \right) \\ &= x^2 \cos x + \frac{2}{x} \bar{y}' \end{aligned}$$

This gives

$$\boxed{u' - \frac{2}{x} u = x^2 \cos x} \quad \text{---(3)}$$

This is a first-order linear differential equation. We know from the theory of exact differential equations that the general solution to (3) is given by

$$\textcircled{1} \quad u = e^{-\int \left(-\frac{2}{x}\right) dx} \left[\int e^{\int -\frac{2}{x} dx} \cdot x^2 \cos x + C \right]$$

where C is an arbitrary constant. Thus,

$$u = e^{\ln x^2} \left[\int e^{\ln \bar{x}^2} \cdot x^2 \cos x dx + C \right]$$

$$= x^2 \left[\int \bar{x}^2 \cdot x^2 \cos x dx + C \right]$$

$$= x^2 (\sin x + C)$$

Resubstitute the value of $y = \bar{u}'$ into the above equation to get

$$\textcircled{1} \quad \boxed{\frac{1}{y} = x^2 (\sin x + C)} \quad - \textcircled{4}$$

Since $y(1) = 2$ we get

①

$$\frac{1}{2} = \sin 1 + C,$$

giving

$$C = \frac{1}{2} - \sin 1$$

The solution to the IVP is given by

$$y = x^{-2} (\sin x + \pi^{-2})^{-1}$$

Consider the initial value problem:

$$\frac{dy}{dx} = \frac{y}{1+x^2}, \quad y(0) = 1$$

- (a) Show that $f(x, y) = \frac{y}{1+x^2}$ satisfies the Lipschitz condition in y on the rectangle

$$R = \{(x, y) \in \mathbb{R}^2 : |x| \leq 1, |y| \leq 2\}.$$

- (b) Use Picard's theorem to justify whether a unique solution exists near $x = 0$. If yes, then find an interval around $x = 0$ where the solution is guaranteed to exist.
- (c) Starting with $y_0(x) = 1$, compute the first two iterations of Picard's method.

Solution and Marking Scheme

- (a) Let $f(x, y) = \frac{y}{1+x^2}$. We compute

$$|f(x, y_1) - f(x, y_2)| = \left| \frac{y_1 - y_2}{1+x^2} \right| \leq |y_1 - y_2|$$

since $1/(1+x^2) \leq 1$ for all x . Therefore, $f(x, y)$ is Lipschitz in y on R with Lipschitz constant $L = 1$.

- **[1 mark]** — Correct use of the definition of Lipschitz condition and appropriate bound.
- (b) The function $f(x, y) = \frac{y}{1+x^2}$ is continuous in both variables, and we have already shown in part (a) that it satisfies Lipschitz condition in y on the rectangle $R = \{(x, y) : |x| \leq 1, |y| \leq 2\}$. The initial point $(0, 1)$ lies in the interior of R . Therefore, we can apply Picard's theorem by choosing a smaller closed rectangle $\tilde{R} = \{(x, y) : |x| \leq a, |y - 1| \leq b\}$ centered at $(0, 1)$ and contained within R . On $\tilde{R}$, the function $f(x, y)$ remains continuous and Lipschitz in y . Let M be a bound for $|f(x, y)|$ on $\tilde{R}$. Since $|y| \leq 1 + b$ and $1 + x^2 \geq 1$, we have

$$|f(x, y)| = \left| \frac{y}{1+x^2} \right| \leq 1 + b =: M.$$

Then Picard's theorem guarantees a unique solution exists on the interval $|x| \leq h$, where $h \leq \min(a, \frac{b}{M})$. For example, taking $a = b = 1$ gives $h \leq \min(1, \frac{1}{2}) = \frac{1}{2}$. Hence, a unique solution is guaranteed to exist on the interval $|x| \leq \frac{1}{2}$.

- **[2 marks]** — Correct identification of continuity and Lipschitz condition and proper explanation of applying Picard's theorem.
- **[1 mark]** — Correct bound computation and conclusion about the interval of existence.

(c) The Picard iteration method uses the recursive formula:

$$y_{n+1}(x) = y_0 + \int_0^x f(t, y_n(t)) dt.$$

[1 mark]

Start with $y_0(x) = 1$.

First iteration:

$$y_1(x) = 1 + \int_0^x \frac{1}{1+t^2} dt = 1 + \tan^{-1}(x).$$

[1 mark]

Second iteration:

$$y_2(x) = 1 + \int_0^x \frac{1 + \tan^{-1}(t)}{1+t^2} dt = 1 + \tan^{-1}(x) + \int_0^x \frac{\tan^{-1}(t)}{1+t^2} dt = 1 + \tan^{-1}(x) + \frac{1}{2}(\tan^{-1}(x))^2.$$

[1 mark]

$$\underline{\text{Ex}} \quad x^3 y'' - 2xy = 6 \ln x \quad x > 0$$

$$\Rightarrow x^2 y'' - 2y = 6 \frac{\ln x}{x}$$

$$x = e^t$$

$$xD = D_1$$

$$x^2 D^2 = D_1(D_1 - 1)$$

$$Dy = \frac{dy}{dx}$$

$$D_1 y = \frac{dy}{dt}$$

$$(D_1^2 - D_1 - 2)y = \frac{6t e^{-t}}{e^t}$$

$$\frac{d^2 y}{dt^2} - \frac{dy}{dt} - 2y = 6t e^{-t} \quad \text{--- (1)}$$

Characteristic equation $r^2 - r - 2 = 0$

has root 2 and -1.

$$\text{C.F.} \quad C_1 e^{2t} + C_2 e^{-t}$$

$$= C_1 x^2 + C_2 \frac{1}{x}$$

(2)

$$\text{P.I.} \quad y_p = \cancel{Ax+B} (At+B) e^{-t}$$

plugging this expression in * we find.

$$2A - 3B = 0, \quad -6A = 6$$

$$\Rightarrow A = -1, \quad B = \frac{2}{3}$$

$$\Rightarrow y_p = \left(-t + \frac{2}{3}\right) t e^{-t}$$

$$= \left(-\frac{2}{3} - \ln x\right) \frac{\ln x}{x}$$

(02)

$$= -\frac{2}{3} \frac{\ln x}{x} - \frac{(\ln x)^2}{x}$$

general solution

$$C_1 x^2 + \frac{C_3}{x} - \left(\frac{2}{3} + \ln x\right) \frac{\ln x}{x}$$

6b

$$y'' - 5y' + 6y = 2e^x$$

characteristic equation

$$r^2 - 5r + 6 = 0$$

roots are 3, 2

(01)

$$C.F. = C_1 e^{3x} + C_2 e^{2x}$$

$$\begin{cases} y_1 = e^{3x} \\ y_2 = e^{2x} \end{cases}$$

$$\begin{aligned} \text{Wronskian } W(y_1, y_2) &= \begin{vmatrix} e^{3x} & e^{2x} \\ 3e^{3x} & 2e^{2x} \end{vmatrix} \\ &= 2e^{5x} - 3e^{5x} = -e^{5x} \end{aligned}$$

using method of V.P.

general solution

$$u_1(x)y_1 + u_2(x)y_2$$

$$\text{where } u_1(x) = - \int \frac{y_2(x)g(x)}{W(y_1, y_2)} dx + C_1$$

(02)

$$= - \int \frac{e^{2x} \times 2e^x}{-e^{5x}} dx + C_1$$

$$= - \frac{2e^{-2x}}{-1} + C_1$$

$$\begin{aligned} u_2(x) &= \int \frac{y_1(x)g(x)}{W(y_1, y_2)} dx + C_2 = \int \frac{e^{3x} \cdot x}{-e^{5x}} + C_1 \\ &= 2e^{-x} + C_1 \end{aligned}$$

general solution

$$C_1 e^{3x} + C_2 e^{2x} + \frac{3x}{e} \begin{pmatrix} -2x \\ -e \end{pmatrix} + \frac{2x}{e} \begin{pmatrix} -x \\ 2e \end{pmatrix}$$

$$= C_1 e^{3x} + C_2 e^{2x} + e^x$$

Q7(a)

$$z = y^2 + 2f\left(\frac{1}{x} + \ln y\right)$$

$$\frac{\partial z}{\partial x} = p = 2f'\left(\frac{1}{x} + \ln y\right) \cdot \left(-\frac{1}{x^2}\right)$$

$$\Rightarrow -x^2 p = 2f'\left(\frac{1}{x} + \ln y\right) \quad \text{--- (A) (1)}$$

$$\frac{\partial z}{\partial y} = q = 2y + 2f'\left(\frac{1}{x} + \ln y\right) \cdot \frac{1}{y}$$

$$\Rightarrow yq - 2y^2 = 2f'\left(\frac{1}{x} + \ln y\right) \quad \text{--- (B) (1)}$$

From (A) and (B)

$$-x^2 p = yq - 2y^2$$

$$\Rightarrow \boxed{x^2 p + yq = 2y^2} \quad (1)$$

Q7(b) $x^2 p + y^2 q = (x+y)z$

The auxiliary equation is

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{(x+y)z} \quad (1)$$

Consider $\frac{dx}{x^2} = \frac{dy}{y^2} \Rightarrow \frac{1}{x} - \frac{1}{y} = c_1 \quad (1)$

considering $-\frac{1}{x} \cdot dx - \frac{1}{y} dy + \frac{1}{z} dz = 0$, and (1)

integrating, we have

$$\ln \frac{z}{xy} = \ln c_2 \Rightarrow \frac{z}{xy} = c_2 \quad (1)$$

Therefore, the general solution is

$$\boxed{\Phi\left(\frac{1}{x} - \frac{1}{y}, \frac{z}{xy}\right) = 0} \quad (1)$$